EX NO : 01

DATE :

AIM:

This exercise relates to the College data set, which can be found in the file college.csv on the book website.

CODE:

```
import pandas as pd
college = pd.read_csv('College.csv', index_col=0)
print(college.head())
print('Rows and columns:', college.shape)
```

OUTPUT:

```
                           Private  Apps  Accept  Enroll  Top10perc  \
Abilene Christian University     Yes  1660    1232     721         23
Adelphi University               Yes  2186    1924     512         16
Adrian College                   Yes  1428    1097     336         22
Agnes Scott College              Yes   417     349     137         60
Alaska Pacific University        Yes   193     146      55         16

                           Top25perc  F.Undergrad  P.Undergrad  Outstate  \
Abilene Christian University         52         2885          537      7440
Adelphi University                   29         2683         1227     12280
Adrian College                       50         1036           99     11250
Agnes Scott College                  89          510           63     12960
Alaska Pacific University            44          249          869      7560

                           Room.Board  Books  Personal  PhD  Terminal  \
Abilene Christian University       3300    450      2200   70        78
Adelphi University                 6450    750      1500   29        30
Adrian College                     3750    400      1165   53        66
Agnes Scott College                5450    450       875   92        97
Alaska Pacific University          4120    800      1500   76        72

                           S.F.Ratio  perc.alumni  Expend  Grad.Rate
Abilene Christian University      18.1           12    7041         60
Adelphi University                12.2           16   10527         56
Adrian College                    12.9           30    8735         54
Agnes Scott College                7.7           37   19016         59
Alaska Pacific University         11.9            2   10922         15
```

OBSERVATION:

The code loads the “college” dataset into a variable called College

B) Look at the data used in the notebook by creating and running a new cell with just the code college in it. You should notice that the first column is just the name of each university in a column named something like Unnamed: 0. We don't really want pandas to treat this as data. However, it may be handy to have these names for later. Try the following commands and similarly look at the resulting data frames:

```
college2 = pd.read_csv('College.csv', index_col=0)
college3 = college.rename({'Unnamed: 0': 'College'},
axis=1)
college3 = college3.set_index('College')
```

This has used the first column in the file as an index for the data frame. This means that pandas has given each row a name corresponding to the appropriate university. Now you should see that the first data column is Private. Note that the names of the colleges appear on the left of the table. We also introduced a new python object above: a dictionary, which is specified by dictionary (key, value) pairs. Keep your modified version of the data withthe following: College = College 3

CODE

```
import pandas as pd
college2 = pd.read_csv('College.csv')
college3 = college2.rename(columns={'Unnamed: 0': 'College'})
college3 = college3.set_index('College')
print(college3.head()
```

OUTPUT

```
                            Private  Apps  Accept  Enroll  Top10perc  \
College
Abilene Christian University     Yes  1660    1232     721         23
Adelphi University               Yes  2186    1924     512         16
Adrian College                   Yes  1428    1097     336         22
Agnes Scott College              Yes   417     349     137         60
Alaska Pacific University        Yes   193     146      55         16

                            Top25perc  F.Undergrad  P.Undergrad  Outstate  \
College
Abilene Christian University         52         2885          537      7440
Adelphi University                   29         2683         1227     12280
Adrian College                       50         1036           99     11250
Agnes Scott College                  89          510           63     12960
Alaska Pacific University            44          249          869      7560

                            Room.Board  Books  Personal  PhD  Terminal  \
College
Abilene Christian University        3300    450      2200   70        78
Adelphi University                  6450    750      1500   29        30
Adrian College                      3750    400      1165   53        66
Agnes Scott College                 5450    450       875   92        97
Alaska Pacific University           4120    800      1500   76        72

                            S.F.Ratio  perc.alumni  Expend  Grad.Rate
...
Adelphi University               12.2           16   10527         56
Adrian College                   12.9           30    8735         54
Agnes Scott College               7.7           37   19016         59
Alaska Pacific University        11.9            2   10922         15
```

OBSERVATION:

The College dataset was successfully loaded using Pandas. The column Unnamed: 0 was renamed to College, and the College column was set as the index. This helps identify each record by its college name and makes the dataset easier to read and analyze.

C) Use the describe() method of to produce a numerical summary of the variables in the data set.

CODE:

```
import pandas as pd
college = pd.read_csv('College.csv', index_col=0)
print(college.describe().round(2))
```

OUTPUT:

```
              Apps    Accept   Enroll  Top10perc  Top25perc  F.Undergrad  \
count    777.00    777.00   777.00     777.00      777.0       777.00
mean    3001.64   2018.80   779.97      27.56       55.8      3699.91
std     3870.20   2451.11   929.18      17.64       19.8      4850.42
min       81.00     72.00    35.00       1.00        9.0       139.00
25%      776.00    604.00   242.00      15.00       41.0       992.00
50%     1558.00   1110.00   434.00      23.00       54.0      1707.00
75%     3624.00   2424.00   902.00      35.00       69.0      4005.00
max    48094.00  26330.00  6392.00      96.00      100.0     31643.00

       P.Undergrad  Outstate  Room.Board    Books  Personal     PhD  Terminal  \
count       777.00    777.00      777.00   777.00    777.00  777.00    777.00
mean        855.30  10440.67     4357.53   549.38   1340.64   72.66     79.70
std        1522.43   4023.02     1096.70   165.11    677.07   16.33     14.72
min           1.00   2340.00     1780.00    96.00    250.00    8.00     24.00
25%          95.00   7320.00     3597.00   470.00    850.00   62.00     71.00
50%         353.00   9990.00     4200.00   500.00   1200.00   75.00     82.00
75%         967.00  12925.00     5050.00   600.00   1700.00   85.00     92.00
max       21836.00  21700.00     8124.00  2340.00   6800.00  103.00    100.00

       S.F.Ratio  perc.alumni    Expend  Grad.Rate
count     777.00       777.00    777.00     777.00
mean       14.09        22.74   9660.17      65.46
std         3.96        12.39   5221.77      17.18
min         2.50         0.00   3186.00      10.00
25%        11.50        13.00   6751.00      53.00
50%        13.60        21.00   8377.00      65.00
75%        16.50        31.00  10830.00      78.00
max        39.80        64.00  56233.00     118.00
```

OBSERVATION:

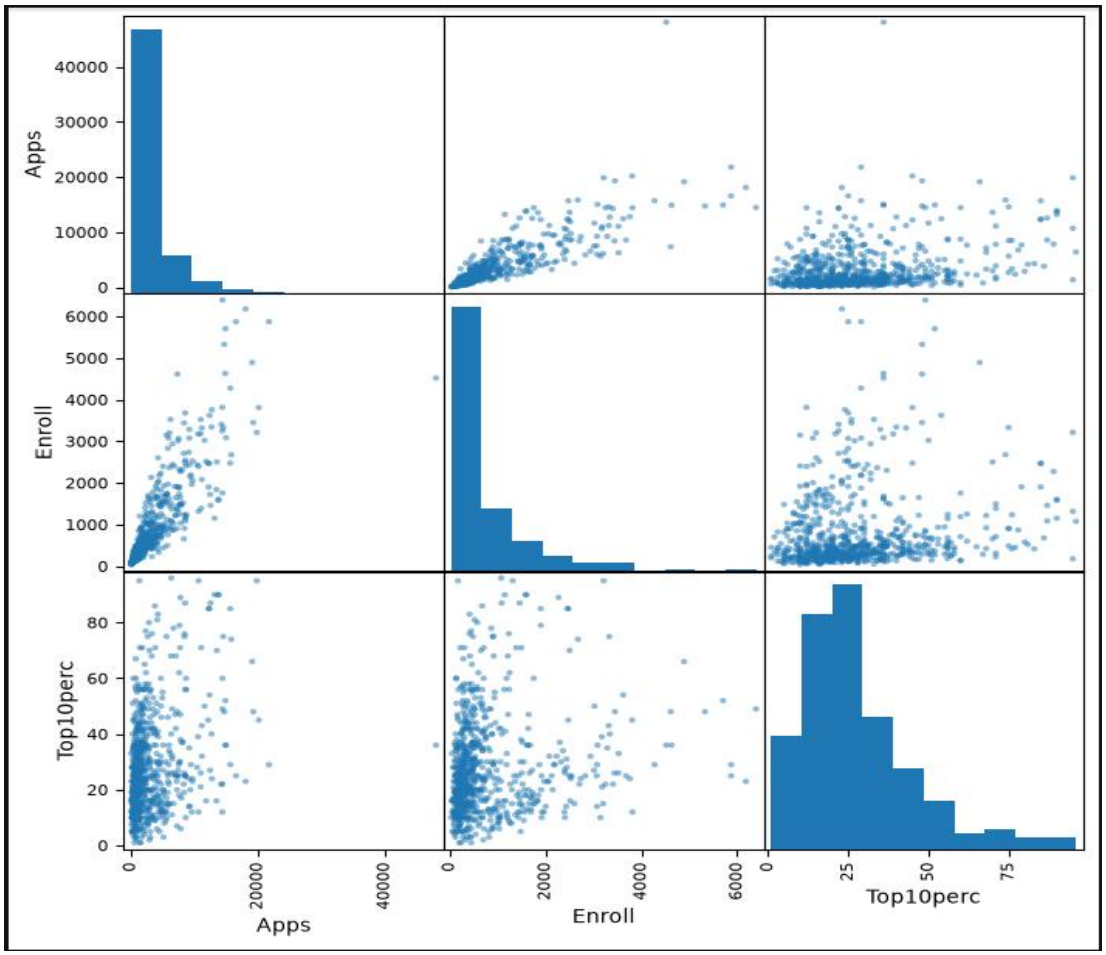
The College dataset contains 777 colleges, with an average of 3,001.64 applications and 779.97 enrolled students per college. The average out-of-state tuition is 10,440.67, and the average graduation rate is 65.46%. The summary statistics show considerable variation among colleges in applications, tuition fees, expenditure, and graduation rates.

D) Use the pd.plotting.scatter_matrix() function to produce a scatterplot matrix of the first columns [Top10perc, Apps, Enroll]. Recall that you can reference a list C of columns of a data frame A using A[C].

CODE

```
import pandas as pd
import matplotlib.pyplot as plt
college = pd.read_csv('College.csv', index_col=0)
pd.plotting.scatter_matrix(college[['Apps', 'Enroll', 'Top10perc']], figsize=(8, 8))
plt.show()
```

OUTPUT



OBSERVATION:

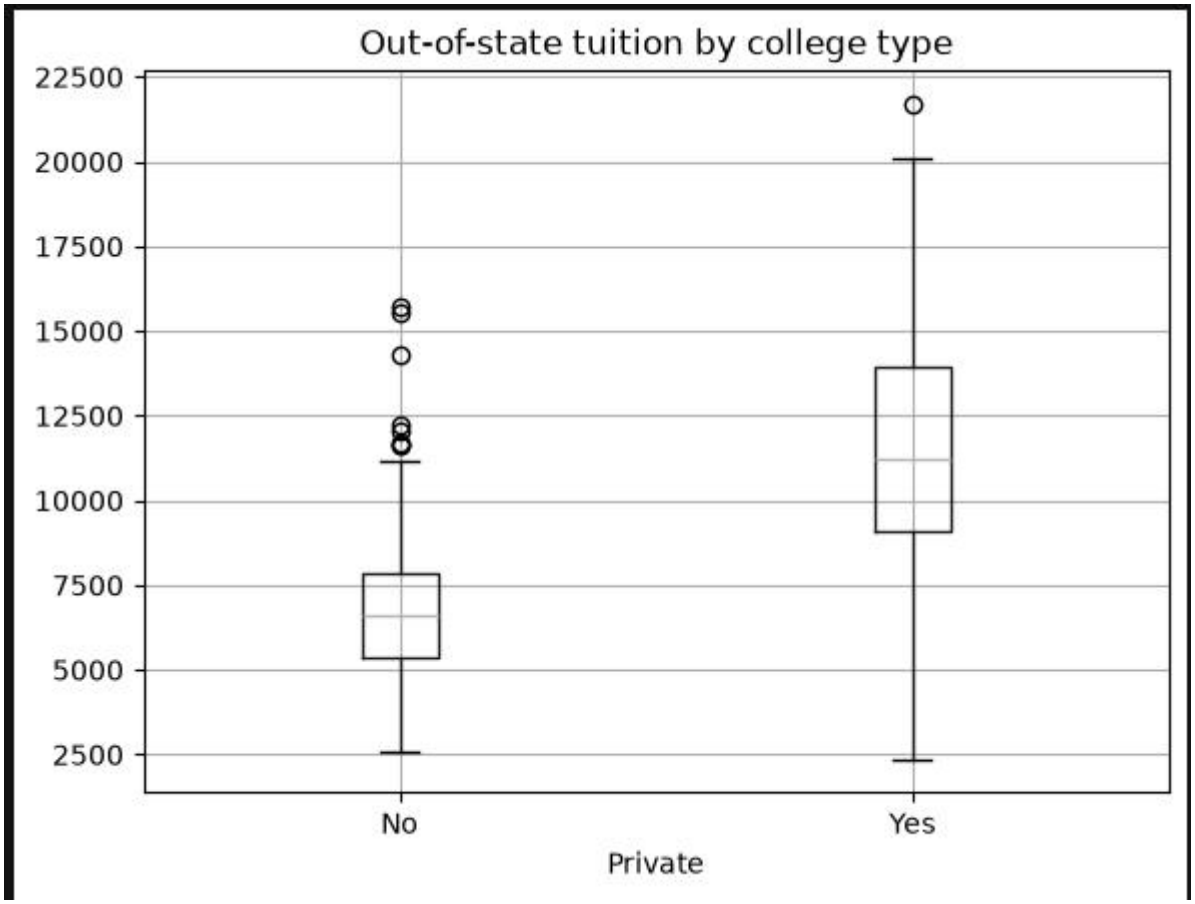
The scatterplot matrix shows a positive relationship between applications (Apps) and enrollment (Enroll), indicating that colleges receiving more applications generally have higher enrollment. The histograms show that most colleges have lower application and enrollment counts. The Top10perc variable is concentrated around lower-to-middle percentages, with a few colleges having very high values.

E) Use the boxplot() method of college to produce side-by-side boxplots of Outstate versus Private.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt
college = pd.read_csv('College.csv', index_col=0)
college.boxplot(column='Outstate', by='Private')
plt.suptitle('')
plt.title('Out-of-state tuition by college type')
plt.show()
```

OUTPUT



OBSERVATION:

The boxplot shows that private colleges generally have higher out-of-state tuition fees than public colleges. Private colleges have a higher median tuition and a wider spread of tuition values. Both groups contain outliers, indicating that some colleges charge considerably higher tuition fees than others.

F) Create a new qualitative variable, called Elite, by binning the Top10perc variable into two groups based on whether or not the proportion of students coming from the top 10% of their high school classes exceeds 50%.

```
college['Elite'] = pd.cut(college['Top10perc'],
[0,0.5,1],
labels=['No', 'Yes'])
```

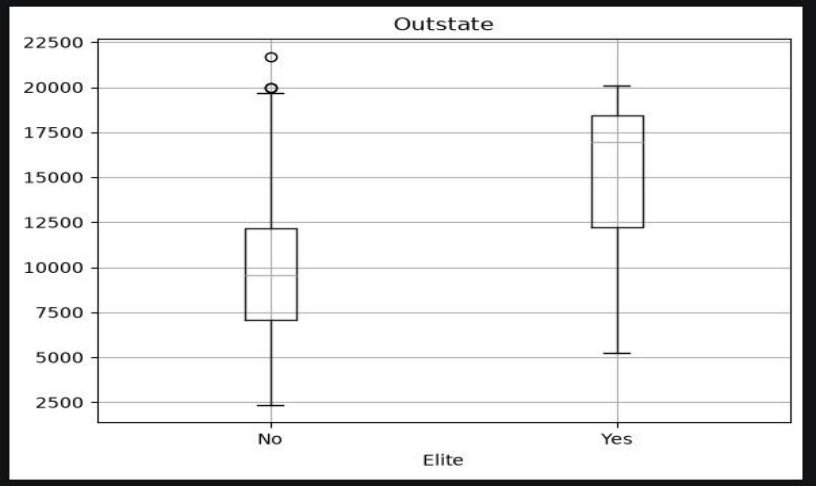
Use the value_counts() method of college['Elite'] to see how many elite universities there are. Finally, use the boxplot() method again to produce side-by-side boxplots of Outstate versus Elite.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt
college = pd.read_csv('College.csv', index_col=0)
college['Elite'] = 'No'
college.loc[college['Top10perc'] > 50, 'Elite'] = 'Yes'
print(college['Elite'].value_counts())
college.boxplot(column='Outstate', by='Elite')
plt.suptitle('')
plt.show()
```

OUTPUT

```
Elite
No     699
Yes     78
Name: count, dtype: int64
```



OBSERVATION:

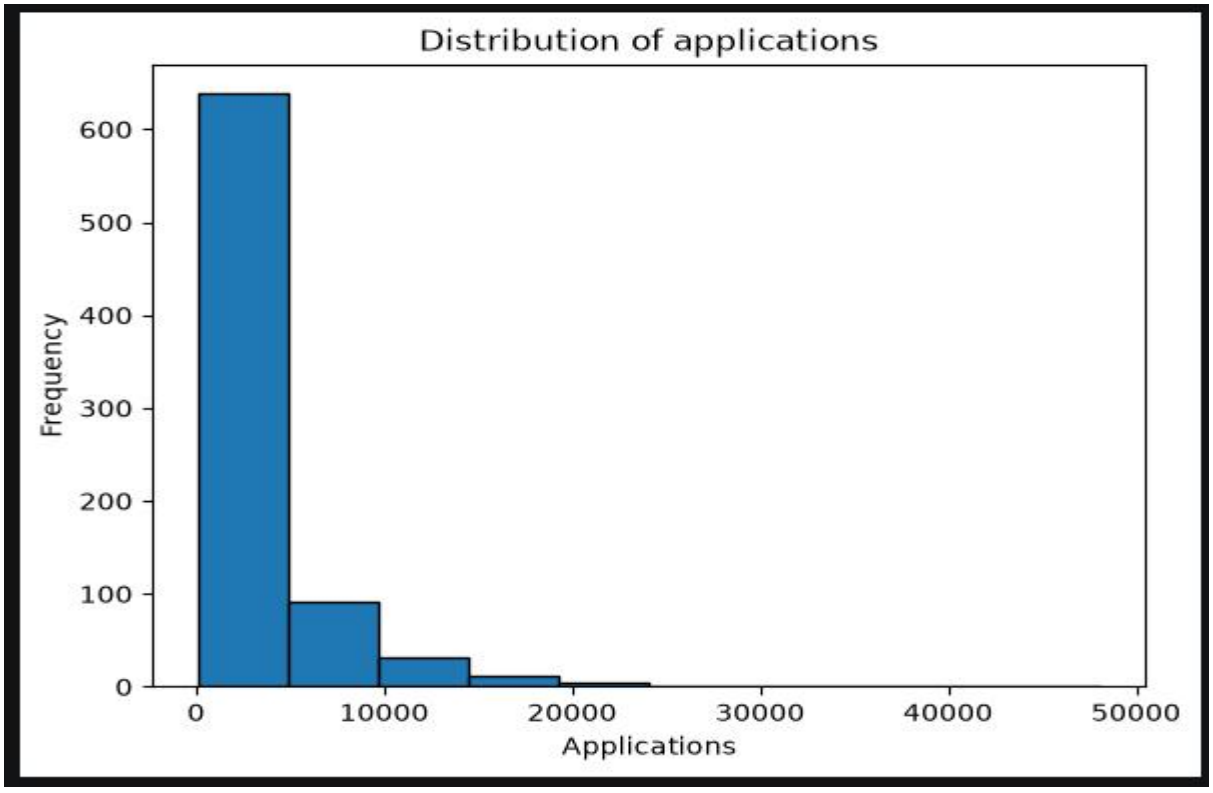
The output shows that 699 colleges are classified as non-elite and 78 colleges as elite. The boxplot indicates that elite colleges generally have higher out-of-state tuition fees than non-elite colleges. The median tuition is higher for elite colleges, suggesting that they tend to charge more for out-of-state students.

G) Use the plot.hist() method of college to produce some his- tograms with differing numbers of bins for a few of the quanti-tative variables. The command plt.subplots(2, 2) may be use- ful: it will divide the plot window into four regions so that four plots can be made simultaneously. By changing the arguments you can divide the screen up in other combinations.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt
college = pd.read_csv('College.csv', index_col=0)
college['Apps'].plot.hist(bins=10, edgecolor='black')
plt.title('Distribution of applications')
plt.xlabel('Applications')
plt.show()
```

OUTPUT



OBSERVATION:

The histogram shows the distribution of applications received by colleges. Most colleges receive fewer than 5,000 applications, while only a small number receive more than 10,000 applications. The distribution is right-skewed, indicating that a few colleges receive a very high number of applications compared to most colleges.

H) Continue exploring the data, and provide a brief summary of what you discover.

CODE

```
import pandas as pd
college = pd.read_csv('College.csv', index_col=0)
print(college.groupby('Private')[['Apps', 'Enroll', 'Outstate']].mean().round(2))
print(college[['Apps', 'Enroll', 'Outstate']].corr().round(2))
```

OUTPUT

```
            Apps   Enroll  Outstate
Private
No       5729.92  1640.87   6813.41
Yes      1977.93   456.95  11801.69
           Apps  Enroll  Outstate
Apps       1.00    0.85      0.05
Enroll     0.85    1.00     -0.16
Outstate   0.05   -0.16      1.00
```

OBSERVATION:

The output shows that public colleges receive more applications and have higher enrollment on average than private colleges. However, private colleges have higher average out-of-state tuition fees (11,801.69) than public colleges (6,813.41). The correlation matrix indicates a strong positive relationship between applications and enrollment (0.85), while out-of-state tuition has a weak relationship with applications (0.05) and a weak negative relationship with enrollment (-0.16).

EX NO : 02

DATE :

AIM:

This exercise involves the Auto data set studied in the lab. Make sure **t**hat the missing values have been removed from the data.

A) Which of the predictors are quantitative, and which are qualitative?

CODE

```
from ISLP import load_data
Auto = load_data('Auto')
print(Auto.dtypes)
print(Auto.head())
```

OUTPUT

```
mpg             float64
cylinders         int64
displacement    float64
horsepower        int64
weight            int64
acceleration    float64
year              int64
origin            int64
dtype: object
                          mpg  cylinders  displacement  horsepower  weight  \
name
chevrolet chevelle malibu  18.0          8         307.0         130    3504
buick skylark 320          15.0          8         350.0         165    3693
plymouth satellite         18.0          8         318.0         150    3436
amc rebel sst              16.0          8         304.0         150    3433
ford torino                17.0          8         302.0         140    3449

                          acceleration  year  origin
name
chevrolet chevelle malibu          12.0    70       1
buick skylark 320                  11.5    70       1
plymouth satellite                 11.0    70       1
amc rebel sst                      12.0    70       1
ford torino                        10.5    70       1
```

OBSERVATION:

The Auto dataset contains information about cars, including mileage (mpg), cylinders, displacement, horsepower, weight, acceleration, year, and origin. The output shows that most columns contain numerical data, while the car name is stored as text. The first five records show cars from model year 1970, with different weights, horsepower, and fuel efficiency values.

B) What is the range of each quantitative predictor? You can an-swer this using the min() and max() methods in numpy. .min()

CODE

```
from ISLP import load_data
Auto = load_data('Auto')
print(Auto.max(numeric_only=True) - Auto.min(numeric_only=True))
```

OUTPUT

```
mpg             37.6
cylinders        5.0
displacement   387.0
horsepower     184.0
weight        3527.0
acceleration    16.8
year            12.0
origin           2.0
dtype: float64
```

OBSERVATION:

The output shows the range of each numerical variable in the Auto dataset, calculated by subtracting the minimum value from the maximum value. The range of mileage (mpg) is 37.6, while weight has the largest range of 3527.0 among the listed variables. These results indicate that cars differ in fuel efficiency, engine characteristics, weight, acceleration, model year, and origin.

C) What is the mean and standard deviation of each quantitative .max()predictor?

CODE

```
from ISLP import load_data
Auto = load_data('Auto')
print(Auto[['mpg','cylinders','displacement','horsepower','weight','acceleration','year']].agg(['mean','std']).round(2))
```

OUTPUT

	mpg	cylinders	displacement	horsepower	weight	acceleration	year
mean	23.45	5.47	194.41	104.47	2977.58	15.54	75.98
std	7.81	1.71	104.64	38.49	849.40	2.76	3.68

OBSERVATION:

The output shows the mean and standard deviation of different variables in the Auto dataset. The average mileage is 23.45 mpg, and the average car weight is 2977.58, with a standard deviation of 849.40. The results indicate that cars vary in fuel efficiency, engine characteristics, weight, and acceleration.

D)Now remove the 10th through 85th observations. What is the range, mean, and standard deviation of each predictor in the subset of the data that remains?

CODE

```
from ISLP import load_data
Auto = load_data('Auto')
Auto_new = Auto.drop(Auto.index[9:85])
print(Auto_new[['mpg','horsepower','weight']].describe().round(2))
```

OUTPUT

	mpg	horsepower	weight
count	281.00	281.00	281.00
mean	25.04	98.72	2881.51
std	7.91	33.82	792.55
min	11.00	46.00	1755.00
25%	18.20	74.00	2188.00
50%	24.50	90.00	2711.00
75%	31.00	112.00	3433.00
max	46.60	230.00	4952.00

OBSERVATION:

After removing observations 10 to 85, the Auto dataset contains 281 cars. The average mileage is 25.04 mpg, average horsepower is 98.72, and average weight is 2882.51. The results show variations in fuel efficiency, horsepower, and weight among the remaining cars.

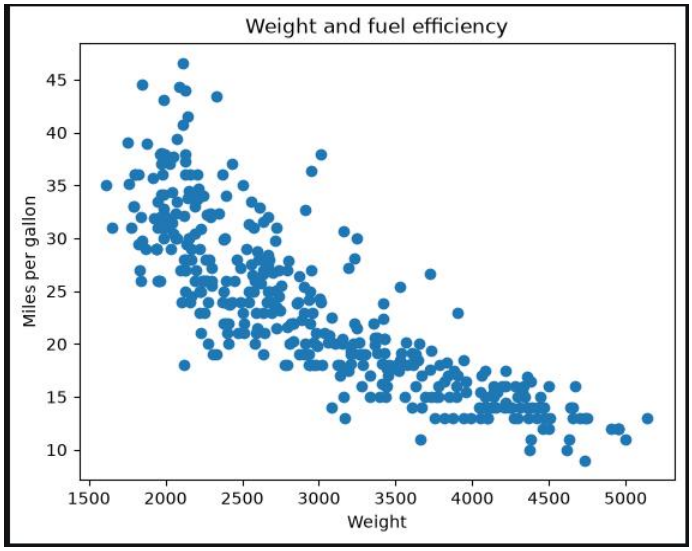
E) Using the full data set, investigate the predictors graphically, using scatterplots or other tools of your choice. Create some plots highlighting the relationships among the predictors. Comment on your findings.

CODE

```
from ISLP import load_data
import matplotlib.pyplot as plt
Auto = load_data('Auto')
plt.scatter(Auto['weight'], Auto['mpg'])
plt.xlabel('Weight')
plt.ylabel('Miles per gallon')
plt.title('Weight and fuel efficiency')
plt.show()
print(Auto[['mpg','weight','horsepower','year']].corr().round(2))
```

OUTPUT

```
              mpg  weight  horsepower  year
mpg          1.00   -0.83       -0.78  0.58
weight      -0.83    1.00        0.86 -0.31
horsepower  -0.78    0.86        1.00 -0.42
year         0.58   -0.31       -0.42  1.00
```



OBSERVATION:

- mpg has a strong negative relationship with weight, displacement and horsepower. Heavier cars with bigger engines get lower mileage.
- The relationship between weight and mpg is slightly curved, not perfectly straight.
- Displacement, horsepower and weight are strongly positively related to each other, so they carry similar information (multicollinearity).
- mpg increases with year, so newer cars tend to have better mileage.
- Cars with 4 cylinders have the highest mpg and cars with 8 cylinders have the lowest.
- Acceleration has only a weak positive relationship with mpg.

F) Suppose that we wish to predict gas mileage (mpg) on the basis of the other variables. Do your plots suggest that any of the other variables might be useful in predicting mpg?

CODE

```
from ISLP import load_data
import statsmodels.formula.api as smf
Auto = load_data('Auto').dropna()
model = smf.ols('mpg ~ weight + horsepower + year', data=Auto).fit()
print(model.summary())
```

OUTPUT

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    mpg   R-squared:                       0.808
Model:                            OLS   Adj. R-squared:                  0.807
Method:                 Least Squares   F-statistic:                     545.4
Date:                Wed, 21 Oct 2026   Prob (F-statistic):          9.37e-139
Time:                        19:12:45   Log-Likelihood:                -1037.4
No. Observations:                 392   AIC:                             2083.
Df Residuals:                     388   BIC:                             2099.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept    -13.7194      4.182     -3.281      0.001     -21.941      -5.498
weight        -0.0064      0.000    -15.768      0.000      -0.007      -0.006
horsepower    -0.0050      0.009     -0.530      0.597      -0.024       0.014
year           0.7487      0.052     14.365      0.000       0.646       0.851
==============================================================================
Omnibus:                       41.952   Durbin-Watson:                   1.227
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               69.490
Skew:                           0.671   Prob(JB):                     8.14e-16
Kurtosis:                       4.566   Cond. No.                     7.48e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 7.48e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

OBSERVATION:

The multiple linear regression model shows that weight, horsepower, and year are used to predict car mileage (mpg). The R-squared value of 0.808 indicates that the model explains approximately 80.8% of the variation in mileage. Weight has a significant negative relationship with mileage, while year has a significant positive relationship; horsepower is not statistically significant because its p-value is 0.597.